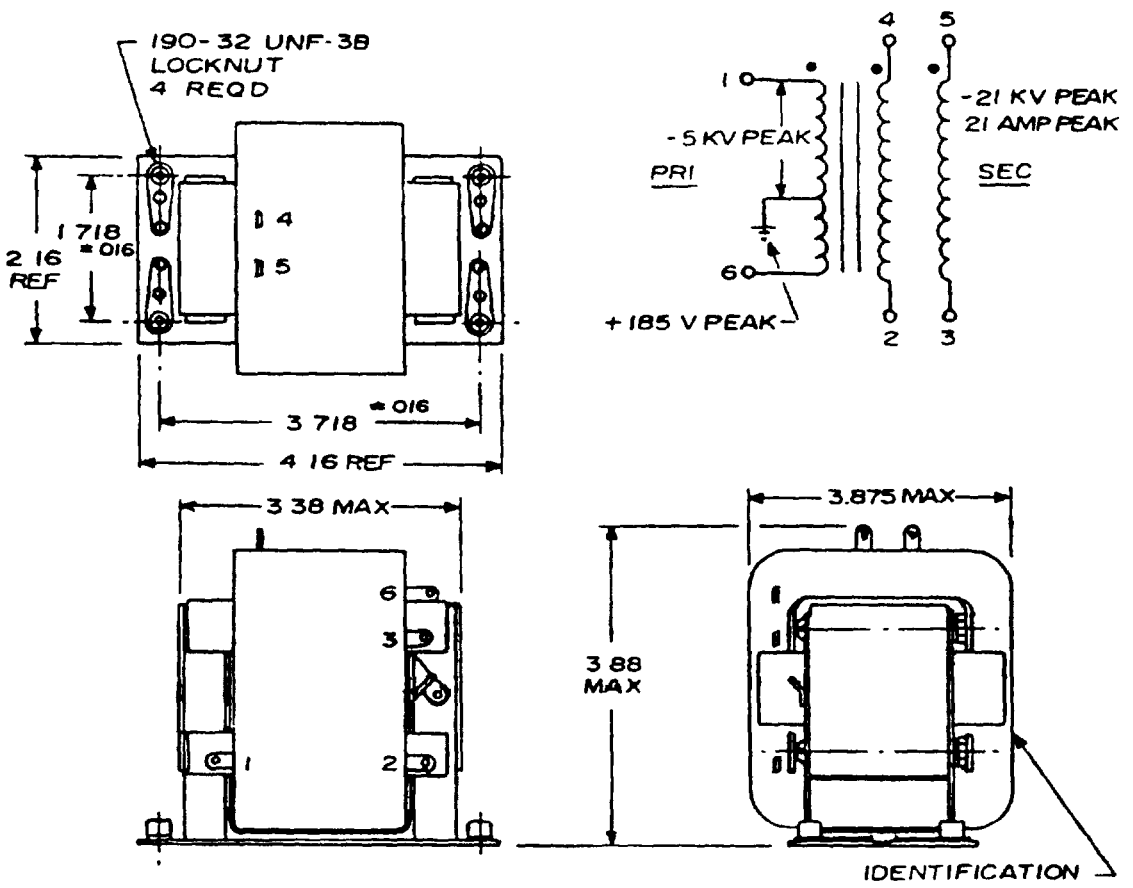
MILITARY SPECIFICATION SHEET

TRANSFORMER, PULSE

The complete requirements for procuring the transformer herein shall consist of this document and the issue in effect of MIL-T-83720.



INCHES	MM	INCHES	MM
.016	.41	3 718	94 44
1 718	43.64	3 875	98 43
2 16	54 86	3 88	98 55
3 38	85 85	4 16	105 66

NOTES
1. DIMENSIONS ARE IN INCHES
2. IDENTIFY TERMINALS AS SHOWN
3. MOUNTING SURFACES, MOUNTING THREADS AND TERMINALS MUST BE FREE OF COMPOUND

FIGURE 1 FSC 5950

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REQUIREMENTS.

Mechanical and Environmental Requirements.

Dimensions and Configuration	See Figure 1
Shock	Test condition A, Method 205
Vibration	Test condition A, Method 204 Amplitude .036 in or 5 G's acceleration
Altitude	10,000 feet
Ambient Temperature Range	-54°C to 95°C
Maximum operating temperature	150°C
Maximum temperature rise	55°C
Weight	Approximately 3.75 lbs
Terminals	Solder type to accomodate 2 leads of No. 18 AWG wire
Life	10,000 hours

Electrical Requirements.

Duty Cycle	.1% max
Voltage Ratio	4.70:1
Source Impedence	50 ohms
Load Impedence	1000 ohms
Load	Magnetron
Pulse Duration	.5 to 1.0 microseconds
Pulse Repetition Rate	1000 PPS
Pulse Peak Power	500 KVA
Primary Input Current	100 amps peak
Pulse Primary Voltage (1-GRD)	5000 V Peak
Pulse Primary Voltage (6-GRD)	185 V Peak

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Pulse Secondary Voltage (4-2 & 5-3)	21,000 V Peak
Pulse Rise Time 20 - 75%	.20 microseconds max
Pulse Fall Time Amplitude	.40 microseconds max
Pulse Droop	10% Max
Exciting Current	75 MA, max with 5V applied to 3 & 5 all other winding open
DC Resistance Windings (2-4) & (3.5)	1 ohm ± 2% max each
Polarity	Dot on schematic indicates same polarity
Bifilar Windings Balance	± max unbalance between windings (4-2) & (5-3) 300 volts
Bifilar Windings	to carry 3.6 amps, 320-1000 Hz filament current
Induced voltage	Apply 6.67 KV, .5 Microsecond pulses to (1-GRD) For 60,000 pulses. Ground terminals (2 & 3). Secondaries terminals (4-5) connected in parallel. Parellel combination connected to 2000 ohm load

Dielectric Withstanding Voltage.

Terminal 4 to Terminal 5	540 V
Terminal 4 & 5 to ground	3000 V
Terminal 4 & 5 to terminal 1	3000 V

Part number: Consists of the basic number of this specification sheet and a dash number.

Example: Part No. M83720/14-001

NOTE: When numbers are used in parenthesis, e.g., (1-2), they indicate the winding and the extreme terminals of the winding.

Custodian:
Air Force - 85

Preparing Activity:
Air Force - 85

Review Activities:
Air Force - 11, 80, 84

(Project 5950-F089-14)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C 16061

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